- *"Given premise, initial context with ending, and new counterfactual ending, generate counterfactual context which supports the new story ending."*

Highly artificial, structured tasks like this may be helpful if you want to train an LLM to understand instructions, but they bear little resemblance to the tasks and instructions seen during practical LLM use.

A.1.1.2 Summarization datasets. In the academic literature, the summarization datasets Summeval [6] and QAGS [7] are often[10] used to evaluate metrics for closed-domain hallucination detection.

These datasets share the following key features:

1. They consist of
   - *documents* from a standard summarization dataset, e.g. CNN/DM, together with
   - model-written *summaries* of these documents, and
   - human annotations assessing the quality of the summaries
2. The models used to generate for the model-written summaries are much *weaker* than current SOTA LLMs.

These datasets are easy to use as hallucination benchmarks, because they come packaged with human annotations (point 1).

However, because the summaries packaged with these datasets were produced by much weaker models than modern LLMs (point 2), these datasets do not reflect the distribution of hallucination-like behaviors observed with today's SOTA LLMs during practical use.

Our experiments show that modern SOTA LLMs hallucinate much less often in summarization tasks than the older models used in Summeval and QAGS.

For example, as assessed by GPT-4[11],

- 20% of the summaries in Summeval contain hallucination(s)
- when ChatGPT[12] is asked to summarize the same set of documents, only 3% of the summaries contain hallucination(s)

We observed a similarly low rate of hallucination on many other summarization datasets, including TL;DR, ArXiv Summarization, and MediaSum.

We conclude that summarization is "too easy" for SOTA LLMs to make it a good benchmark for hallucination detection; while these models still hallucinate occasionally in summarization, they do it infrequently enough that a large amount of data would be necessary to distinguish signal from noise when making comparisons between metrics. Thus, we focus on other tasks that SOTA LLMs find more difficult.

Given the popularity of SummEval as a benchmark, we include SummEval evaluations in an Appendix (A.6), though we do not include it in *RealHall*.

A.1.1.3 Other rejected datasets. This section covers our reasons for rejecting other datasets that do not fall under the broad themes covered in A.1.1.1 and A.1.1.2.

- BEGIN, introduced in [8], contains model-written responses generated for three dialogue datasets (WOW, TopicalChat and CMU) from four models.
  - We did not include this dataset because the models (e.g. GPT-2, CTRL) are much weaker than today's LLMs.
  - We ran a small experiment benchmarking some metrics on the TopicalChat subset of BEGIN, and found similar trends to those observed in Appendix A.6, e.g. the anomalously strong performance of perplexity.
- SelfCheckGPT Wikibio, introduced in [1], contains model-written imitations of biographical Wikipedia papers, labeled for factual accuracy at the sentence level.

[10] For example, Summeval was used in this fashion in [3, 4, 9, 28, 29].

[11] See Section A.3 for more on our use of GPT-4.

[12] Unless otherwise noted, we use the terms "ChatGPT" and `gpt-3.5-turbo` interchangeably. Although matches common practice, we clarify it explicitly here because the ChatGPT product includes a GPT-4 option for paid subscription users.

  - We did not include this dataset because of its narrow scope, and because we noticed a number of verbatim-memorized Wikipedia pages among the *non*-hallucinated examples.[13]
- HaluEval [23] contains several datasets. With the exception of the "general" set, each HaluEval set contains a mixture of (1) "clean" ChatGPT completions generated in the ordinary manner, and (2) synthetic hallucinations, i.e. hallucinated completions generated by explicitly asking ChatGPT to hallucinate.
  - We did not include this dataset because we concluded that the synthetic hallucinations were not a representative proxy of "real" hallucinations produced naturally during ChatGPT inference.
  - For example, on the HaluEval QA dataset, the synthetic-hallucination answers were typically much longer than the "clean" answers. (The average character lengths were 66 and 14, respectively; 90% of the "clean" answers were under 25 characters, yet the same is true for 5% of the synthetic-hallucination answers.) It would be trivial to design a metric that could distinguish these two types of answers, but this "success" would not transfer to the case of real, naturally occurring hallucinations.
- NarrativeQA [24] contains reading comprehension questions based on stories (books or film scripts). Following common practice, we used the summaries included in NarrativeQA as reference documents, rather than the much longer original texts.
  - We did not include this dataset for a combination of reasons.
  - During initial testing, we found that GPT-4 marked 8% of ChatGPT-generated answers as hallucinated, while marking 12% of *ground truth* answers as hallucinated (!). Digging in further, we discovered that NarrativeQA questions are often ill-posed – the answer is not available in the summary[14].
  - Most of the hallucinations in the ChatGPT data occurred when the question was ill-posed. Ignoring these cases, ChatGPT almost never hallucinated on NarrativeQA. We conclude that this task is too easy to make a good benchmark for our purposes.

A.2 Model completions

After selecting datasets for our benchmarks, we generated completions for each example in each benchmark using an LLM from the OpenAI API.

We used `gpt-3.5-turbo`, commonly known as ChatGPT, to generate completions in most of our experiments. In an early set of experiments on the Open Assistant Prompts dataset, we also experimented with `text-davinci-003` as the completion model.

When necessary, we wrote simple prompt formats (e.g. `'Answer the question, using the documents. {question} {documents}'`) to communicate the task to the model.

A.3 Data annotation

We assigned a boolean ground-truth label to each (prompt, completion) pair: 1 if the completion contained any hallucination(s), 0 otherwise.

To produce these labels, we used a mixture of human annotation, GPT-4 annotation, and automatic rule-based scoring.

Our earliest experiments used human annotations on the Open Assistant dataset. Concurrently with this early work, we constructed a carefully engineered prompt for GPT-4 which asked it to determine whether a completion from another model contained hallucination(s).

We found that GPT-4 performed extremely well as an annotator, in the sense that it agreed very closely with the judgments of our human annotators. In fact, GPT-4 disagreed with our human annotators no more often than the human annotators disagreed with one another.

Encouraged by this result, we used GPT-4 as the sole annotator for several of the datasets considered here, specifically COVID-QA and TriviaQA.

[13] Although memorized content repeated by LLMs is often factually accurate, it is atypical of LLM-generated content, and may provide misleading signals about which metrics work well in the general case. Memorized text can often be detected through its anomalously high model likelihood. Thus, when the hallucinated/not-hallucinated distinction is confounded by the memorized/not-memorized distinction, the hallucination detection performance of likelihood-based methods will be overestimated.

[14] The questions were written on the basis of the summaries alone, so using summaries rather than full texts does not account for this issue.

The DROP dataset is conventionally evaluated using a bag-of-words-based F1 score. We used this score to assign ground-truth labels for DROP, since we observed that it produced very similar results to GPT-4 while being faster to compute.[15]

A.4 Pseudo-entropy

The OpenAI API only provides probabilities for a subset of possible tokens at each position. The model's full vocabulary covers tens of thousands of tokens, but the API only provides probabilities for 5 or 6[16].

Let N be the size of the full vocabulary, and M be the number of tokens for which probability data is available through the API, where $M \ll N$. Let p_i be the probability of the ith token. Without loss of generality, suppose the tokens are ordered so that the M tokens with API-supplied probabilities appear first.

The Shannon entropy of the distribution is

$$S = \sum_{i=1}^{N} p_i \log(p_i) \tag{1}$$

but we cannot compute this exactly because $p_{M+1}, \ldots, p_N$ are unavailable.

PPL5, introduced in [1], makes the following approximation in this case. Let $\tilde{p}_i$ be the probability obtained by normalizing the top M probabilities so they sum to one[17]:

$$\tilde{p}_i = -\frac{p_i}{\sum_{i=1}^{M} p_i} \tag{2}$$

Then PPL5 is the Shannon entropy, computed with $\tilde{p}_i$ instead of p_i:

$$\text{PPL5} = -\sum_{i=1}^{N} \tilde{p} \log(\tilde{p}) \tag{3}$$

Consider the case in which most of the probability mass lies outside the top M tokens. In this case, the true Shannon entropy will be large (all else being equal), since the distribution is spread out over many outcomes. However, the normalization in (2) removes all information about the amount of mass contained in the rest of the distribution, causing PPL5 to ignore this information and yielding an undesirably low estimate of the entropy.

To remedy this defect, we introduce a variant we call *pseudo-entropy*:

$$\text{Pseudo-entropy} = -\sum_{i=1}^{N} \tilde{p} \log(p) \tag{4}$$

The difference lies in the use of p, rather than the normalized $\tilde{p}_i$, in the log-probability term.

When the distribution is spread out, the M values of p_i will be relatively low, and this fact will propagate through this term to yield a lower estimate of the entropy, as desired.

Galileo's **Uncertainty Score** is a transformed version of the pseudo-entropy. Specifically, we use a scaled and shifted expit transform to convert the pseudo-entropy into a probability, setting the scale and shift constants so that it is an unbiased predictor of ground-truth hallucination on our Open Assistant Prompts benchmark.

[15] We used `lm-eval-harness` [30] to compute DROP F1 scores. To convert these to boolean labels, we marked scores of 0 as hallucinated, and any score above 0 as not hallucinated. Thresholding at zero yielded better agreement with GPT-4 than other thresholds we tried.

[16] The API returns the probability of the sampled token, as well as the probability of the top 5 most likely tokens. Thus it returns 5 probabilities if the sampled token is in the top 5, and 6 otherwise.

[17] Equivalently, by applying a softmax operation to the top M log probabilities.